Homework-3 solutions

CS224W (Fall 2021)
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Question 1: GraphRNN

Q1.1

The sequence is

$$\{S^\pi_A : [], S^\pi_B : [S^\pi_{B,A} = 1], S^\pi_D : [S^\pi_{D,A} = 1, S^\pi_{D,B} = 0], S^\pi_C : [S^\pi_{C,A} = 0, S^\pi_{C,B} = 1, S^\pi_{C,D} = 0],$$
$$S^\pi_E : [S^\pi_{E,A} = 0, S^\pi_{E,B} = 1, S^\pi_{E,C} = 0, S^\pi_{E,D} = 1], S^\pi_F : [S^\pi_{F,B} = 0, S^\pi_{F,C} = 1,$$
$$S^\pi_{F,D} = 0, S^\pi_{F,E} = 1]\}$$

Note that at Node F, the model is not asked to predict $S^\pi_{F,A}$ because this is 0 by design. This is one of the advantages mentioned in the next question.

Q1.2

There are two primary benefits to this ordering, these are from the paper but with my intuition added,

1. Firstly, we have destroyed the underlying graph structure and are modeling it as a sequence. In order for the model to learn an invariant representation to node-ordering we would need to train it on all permutations (this is a type of augmentation). If we use a random ordering, regardless of the structure we

*I solved them while I was an undergraduate at BITS Pilani, Goa Campus.

have $\mathcal{O}(n!)$ permutations, instead if we train only on BFS orderings, we have to model much fewer permutations for real-world graphs[1].

2. Secondly, we only need to model edge decisions which are *at most 1 breadth level above*, i.e., like the node F in the previous question couldn't be connected to node A. It took me some time to absorb this fact, the lecture definitely helped, the proofs in [**?**] do too (might be complicated if you're encountering BFS for the first time). Considering the previous example, if node F were to be connected to node A, then it would have been generated before nodes E and F[2].

Question 2: Subgraphs and Order Embeddings

Q2.1

As B is a subgraph of C,

$$\begin{aligned}&\exists g : V_B \rightarrow V_C \\ &\text{ such that the subgraph of } C \text{ induced by} \\ &\{g(w) | w \in V_B\} \text{ is graph isomorphic to } B.\end{aligned}$$

I will now prove that all subgraphs of B are subgraphs of C using proof by contradiction [3]. Assume $\exists V_H \subseteq V_B$ such that H is a subgraph of B, but the subgraph of C induced by $\{g(w)|w \in V_H\}$ is not graph isomorphic to H.

In this case,

$$\exists u, v \in V_H | (u, v) \in E_H \text{ and } (g(u), g(v)) \notin E_C \tag{1}$$

But by definition of g, we have,

$$\forall u, v \in V_B | (u, v) \in E_B \implies (g(u), g(v)) \in E_C \tag{2}$$

Clearly, 1 contradicts 2. Therefore, H cannot exist, and all subgraphs of B are subgraphs of C. Thus, A is a subgraph of C.

[1] As mentioned in the paper, fully-connected graphs still have $\mathcal{O}(n!)$ BFS orderings.

[2] It helped to think how this would be implemented, we would maintain a queue of 'frontier' nodes; if nodes E and F have been generated and were not connected to node A, this queue wouldn't contain node A.

[3] There is also a constructive proof possible, which shows that the required function is just $g(f(\cdot))$, but I think it would be more involved, so I just didn't go that route.

Q2.2

We are given that A is a subgraph of B, and that B is a subgraph of A. Using the previous definition,

$$\begin{aligned}
&\exists f : V_A \to V_S | (S \subseteq B) \text{ and } f \text{ is bijective} \\
&\exists g : V_B \to V_T | (T \subseteq A) \text{ and } g \text{ is bijective} \\
&\implies |V_A| = |V_S| \leq |V_B| \\
&\implies |V_B| = |V_T| \leq |V_A| \\
&\implies |V_A| = |V_B|
\end{aligned}$$

Finally, we have $S = B$, which means that the subgraph induced by $\{f(w) | w \in V_A\}$ is graph isomorphic to B, thus, A and B are graph isomorphic.

Q2.3

First, I will prove the '$\rightarrow$' direction, i.e., if X is a common subgraph of A and B, then $z_X \preccurlyeq min\{z_A, z_B\}$.

$$\begin{aligned}
&\text{As } X \text{ is a subgraph of } A \\
&z_X \preccurlyeq z_A \\
&\text{Also } X \text{ is a subgraph of } B \\
&z_X \preccurlyeq z_B \\
&\implies z_X \preccurlyeq min\{z_A, z_B\}
\end{aligned}$$

Now to prove the '$\leftarrow$' direction, i.e., if $z_X \preccurlyeq min\{z_A, z_B\}$, then X is a common subgraph of A and B.

$$\begin{aligned}
&z_X \preccurlyeq min\{z_A, z_B\} \\
&\implies z_X \preccurlyeq z_A \text{ and } z_X \preccurlyeq z_B \\
&\implies X \text{ is a subgraph of } A \text{ and } X \text{ is a subgraph of } B \\
&\implies X \text{ is a common subgraph of } A \text{ and } B
\end{aligned}$$

Q2.4

The ordering is

$$z_C[1] > z_B[1] > z_A[1]$$

if any of these were not true, say for example, that $z_B[1] \leq z_A[1]$, then B would be a subgraph of A, which is not true as given in the question. Similar arguments hold for other inequalities.

Q2.5

For simplicity, I assume [4],

$$z_C[1] > z_B[1] > z_A[1] \tag{3}$$
$$z_A[0] > z_B[0] > z_C[0] \tag{4}$$

I will present X, Y and Z such that,

$$X \text{ is a common subgraph of } A, B, C$$
$$Y \text{ is a common subgraph of } B, C$$
$$Z \text{ is a common subgraph of } A, B$$

The counterexample is shown in Fig. 1.

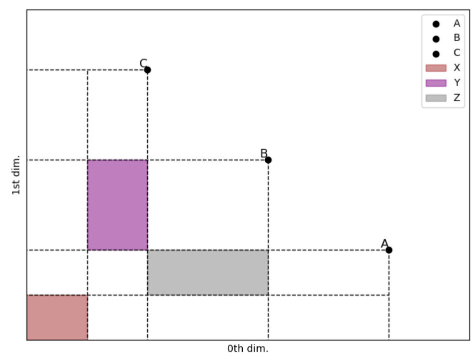

Figure 1: Counterexample for Q2.5, X can be anywhere in the highlighted region, similar for Y and Z.

Explictly, for graph X,

$$z_X[0] \leq z_C[0]/2$$
$$z_X[1] \leq z_A[1]/2$$

[4]We can do this because in Q2.4, we derived this result *without loss of generality*, i.e., results derived from A, B, C obeying this condition would hold for general non-isomorphic A, B, C.

for graph Y,

$$\begin{aligned}
z_C[0]/2 < z_Y[0] &\leq z_C[0] \\
z_Y[1] &\leq z_B[1] \\
z_Y[1] &> z_A[1]
\end{aligned}$$

for graph Z,

$$\begin{aligned}
z_Z[0] &\leq z_B[0] \\
z_Z[0] &> z_C[0] \\
z_A[1]/2 < z_Z[1] &\leq z_A[1]
\end{aligned}$$

Clearly, we have $z_Z \preccurlyeq z_Y$ and $z_Z \preccurlyeq z_X$.